



# **Walchand College Of Engineering, Sangli.**

**(An Autonomous Institute)**

**Department  
of  
Computer Science and Engineering**

**Mini-Project-2 (7VSCS346)  
Synopsis  
on**

## **Privacy-Preserving Hybrid LLM System for Secure Document Reasoning**

by

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2025-26

## 1. Problem statement

Many organizations such as hospitals, banks, law firms, and corporate enterprises handle highly sensitive documents. Cloud-based Large Language Models (LLMs) provide strong reasoning capabilities but cannot be used directly due to privacy regulations and data confidentiality risks. Purely local models preserve privacy but often lack strong multi-step reasoning ability.

To design and develop a privacy-preserving AI system that improves reasoning quality of local models without transmitting sensitive data to the cloud.

## 2. Abstract

This project proposes a Privacy-Preserving Hybrid LLM Architecture for secure document reasoning. The system integrates a local Large Language Model (LLM) with a cloud-based reasoning planner. Sensitive documents remain on the local machine, while only abstract task descriptions are sent to the cloud for structured reasoning plan generation.

The cloud model generates a step-by-step reasoning plan without accessing private data. The local model executes the plan using retrieved document content through a Retrieval-Augmented Generation (RAG) pipeline. The system includes a query abstraction module to prevent data leakage and a comparison framework to evaluate local-only versus hybrid versus cloud reasoning performance.

The final output is a web-based prototype demonstrating improved reasoning quality while maintaining strict privacy constraints.

## 3. Problem Domain

- Artificial Intelligence
- Natural Language Processing
- Privacy Preserving AI Systems
- Retrieval-Augmented Generation (RAG)

## 4. Project Area mapped to relevant SGD

SDG 4 – Quality Education

(Development of privacy-preserving AI systems for secure learning and academic research)

SDG 9 – Industry, Innovation and Infrastructure

(Development of secure AI infrastructure for enterprises)

## 5. Objectives

1. To study Large Language Models (LLMs), their architecture, reasoning capabilities, and security aspects.
2. To design and develop a secure hybrid LLM architecture separating reasoning, planning, and execution layers.
3. To test and compare Local-Only and Hybrid pipelines using evaluation metrics.
4. To develop a user-friendly UI for the proposed hybrid LLM architecture.

6. **Outcomes/ Deliverables** (deployment components such as portal, app, prototype, product etc)

- Working Web Application (Streamlit-based interface)
- Hybrid LLM pipeline (Local + Cloud integration)
- Query Structuring and Privacy Filter Module
- RAG-based local document processing system
- Performance comparison report (Local vs Hybrid vs Cloud)
- Source code repository

7. **Project Potentials** (scope/ benefits, patentability, startup, paper publication)

**Scope / Benefits:**

- Enables AI adoption in privacy sensitive domains.
- Improves reasoning capability of smaller local models.
- Reduces dependency on full cloud processing.

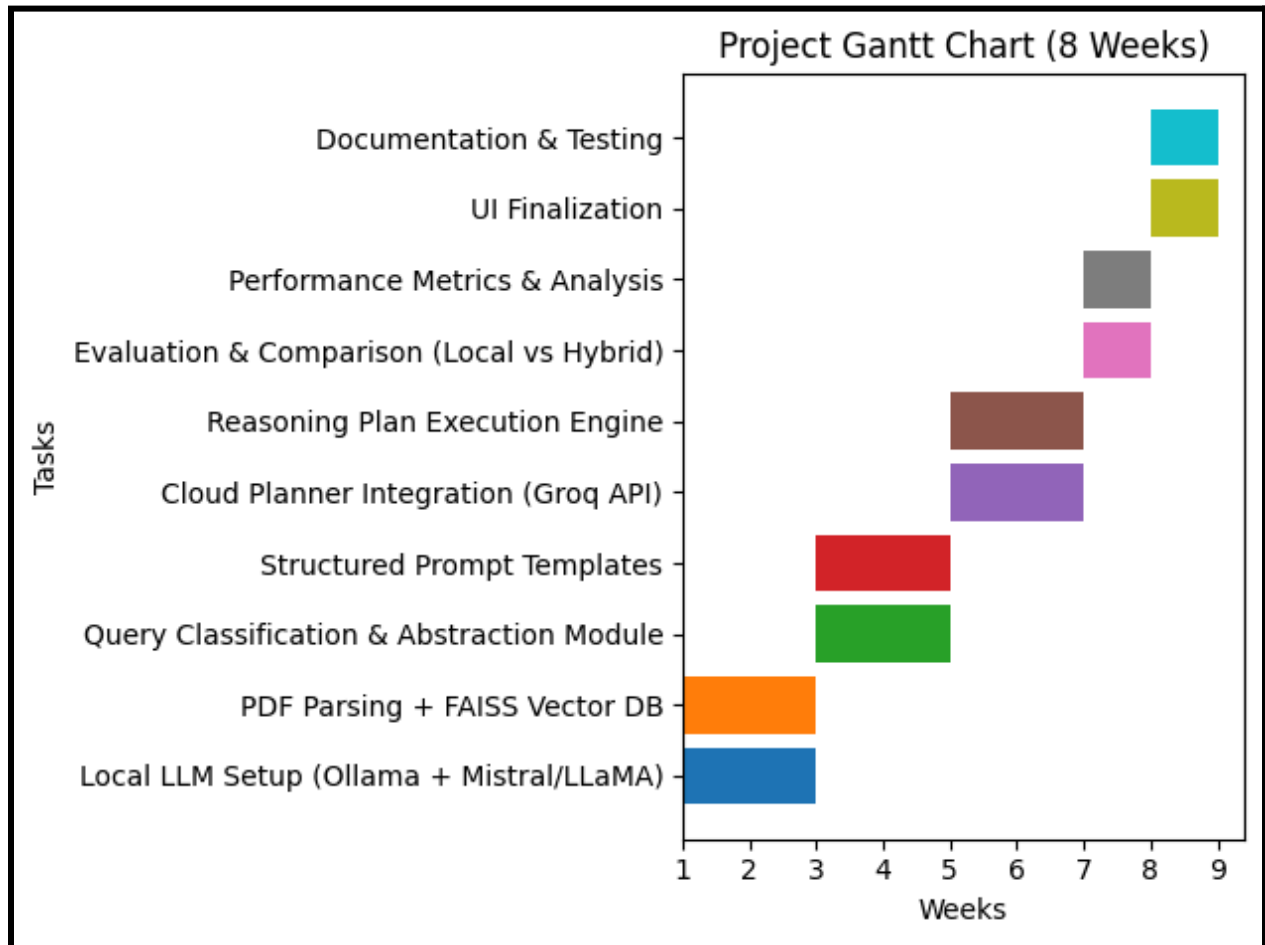
**Startup Potential:**

- Can be developed into an enterprise-grade secure AI assistant.
- Applicable to banking, healthcare, and legal industries.

**Research Publication Potential:**

- Comparative evaluation of reasoning enhancement using hybrid architecture.
- Analysis of privacy-preserving planning-execution separation.

## 8. Project plan



### Week 1-2:

Local LLM setup (Ollama + Mistral/LLaMA)  
PDF parsing and vector database (FAISS)

### Week 3-4:

Design query classification and abstraction module  
Develop structured prompt templates

### Week 5-6:

Integrate cloud planner (Groq API)  
Develop reasoning plan execution engine

### Week 7:

Evaluation and comparison (Local vs Hybrid)  
Performance metrics and analysis

### Week 8:

UI finalization and testing

**Work Distribution:**

**Student 1:**

Local LLM + RAG pipeline + Vector DB

**Student 2:**

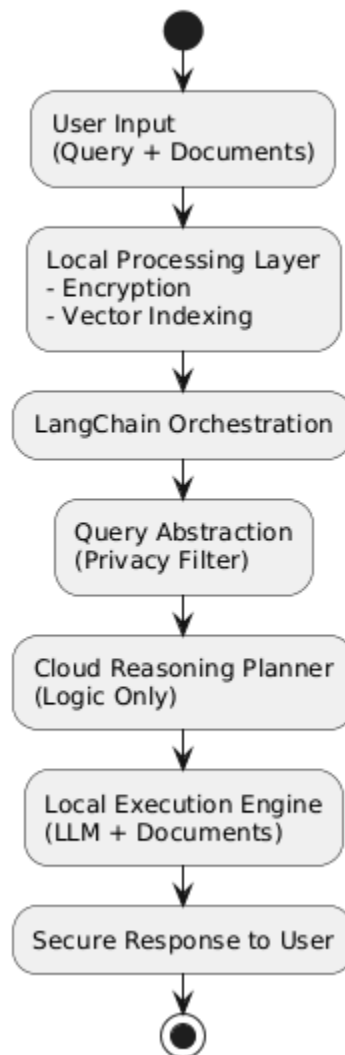
Cloud integration + Planning module + API management

**Student 3:**

Query abstraction + UI + Evaluation metrics + Testing

**9. Methodology** (tech stack, algorithms, techniques, services etc)

**Methodology Overview: Privacy-Preserving Hybrid LLM System**



- The user submits a query and private documents through a web interface.
- Documents are encrypted, indexed, and stored locally.
- LangChain orchestrates the complete workflow.
- The query is abstracted to remove sensitive information.
- A cloud LLM generates a reasoning plan using only abstract logic.
- The local LLM executes the plan on private documents.
- The final secure response is returned to the user.

**Tech Stack:**

- Python
- LangChain (orchestrator)
- Ollama (local LLM execution)
- Cloud Reasoning Planner: Groq API / Together AI
- FAISS (vector database)
- Streamlit (web interface)

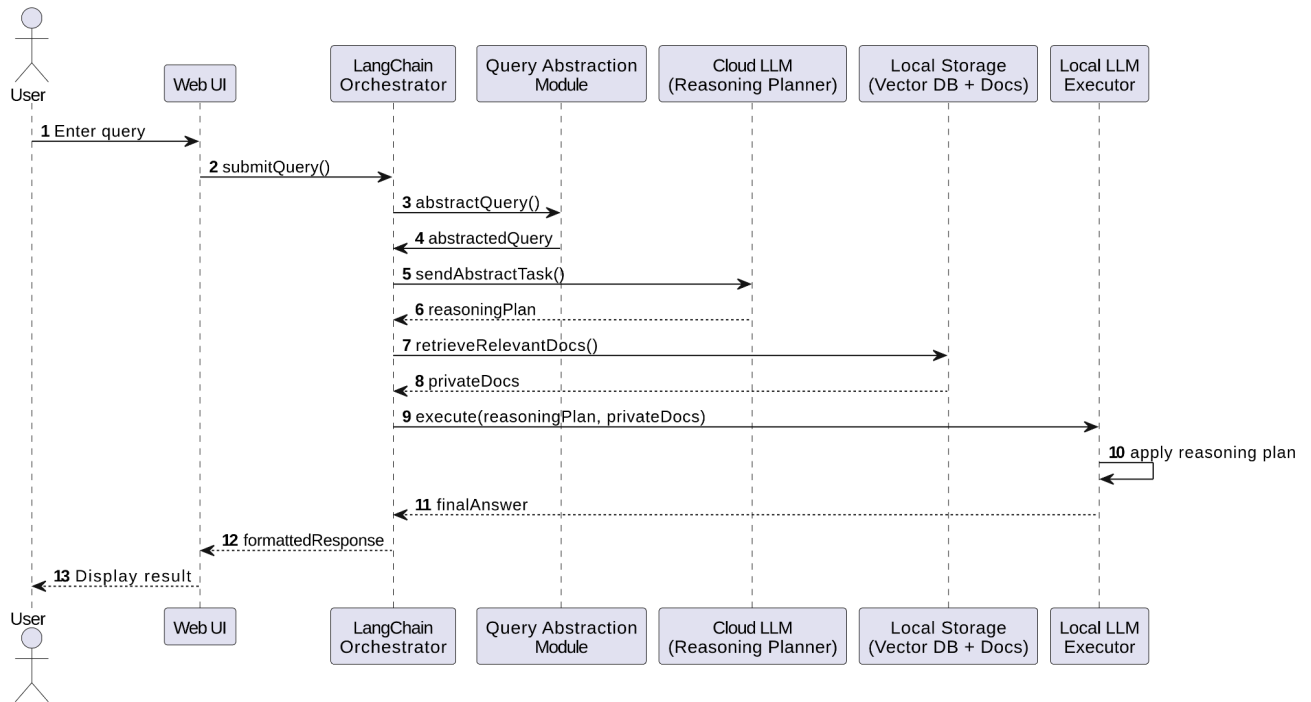
**Techniques Used:**

- Retrieval-Augmented Generation (RAG)
- Query classification and abstraction
- Prompt engineering
- Structured reasoning decomposition
- Planning-Execution architecture separation

**Algorithms:**

- Cosine similarity for document retrieval
- Intent classification for query structuring
- Step-by-step reasoning execution

## 10. UML diagrams



1. The user enters a query through the Web UI.
2. The Web UI forwards the query to the LangChain Orchestrator.
3. LangChain sends the query to the Query Abstraction Module to remove sensitive details.
4. The abstracted query is sent to the Cloud LLM to generate a reasoning plan.
5. The Cloud LLM returns only the reasoning plan, without accessing any private data.
6. LangChain retrieves relevant private documents from local storage.
7. The reasoning plan and documents are sent to the Local LLM for execution.
8. The Local LLM applies the reasoning plan on private documents.
9. The final answer is returned to the Web UI and displayed to the user.

## 11. References

1. Bae, Y., Kim, M., Lee, J., Kim, S., Kim, J., Choi, Y. and Miresghallah, N., 2025. PPMI: Privacy-Preserving LLM Interaction with Socratic Chain-of-Thought Reasoning and Homomorphically Encrypted Vector Databases. arXiv preprint arXiv:2506.17336.
2. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.T., Rocktäschel, T. and Riedel, S., 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. Advances in neural information processing systems, 33, pp.9459-9474.

### Web References

1. <https://docs.langchain.com/>
2. <https://developers.openai.com/api/docs/>